Data-Splitting Techniques in Machine Learning

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1. Introduction

This report evaluates 16 widely used data splitting and resampling techniques on a single dataset, the UCI Heart Failure Prediction dataset (*heart.csv*), using a consistent evaluation setup with a RandomForestClassifier (100 trees, fixed random seed). The main objective is to empirically examine how different data splitting strategies influence the reported performance of the same model on the same dataset, and to highlight the trade-offs in terms of bias, variance, computational cost, and practical suitability.

For each method, the results include train/test or cross-validation accuracy (depending on the approach), along with per-fold summaries where applicable. All techniques are then compared in a consolidated ranking table and visualized using a final comparative bar chart.

Dataset Overview:

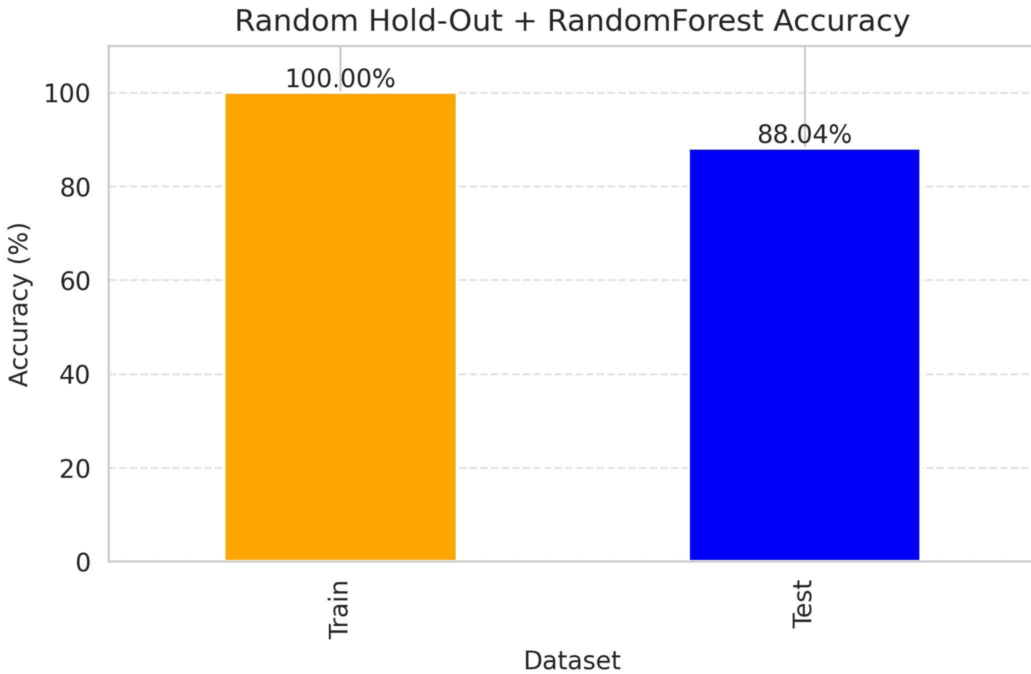
- Source file: heart.csv
- Records: 918 patients
- Features: Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExerciseAngina, Oldpeak, ST_Slope (11 features, categorical columns label-encoded)
- Target: HeartDisease (binary: 0 = no disease, 1 = disease)
- Model: RandomForestClassifier(n_estimators=100, random_state=42)

2. Technique-by-Technique Results

2.1. Random Hold-Out

The most common hold-out approach: records are randomly assigned to Training and Test subsets (80:20 split), with no shuffling discipline made explicit and no class-balance guarantee.

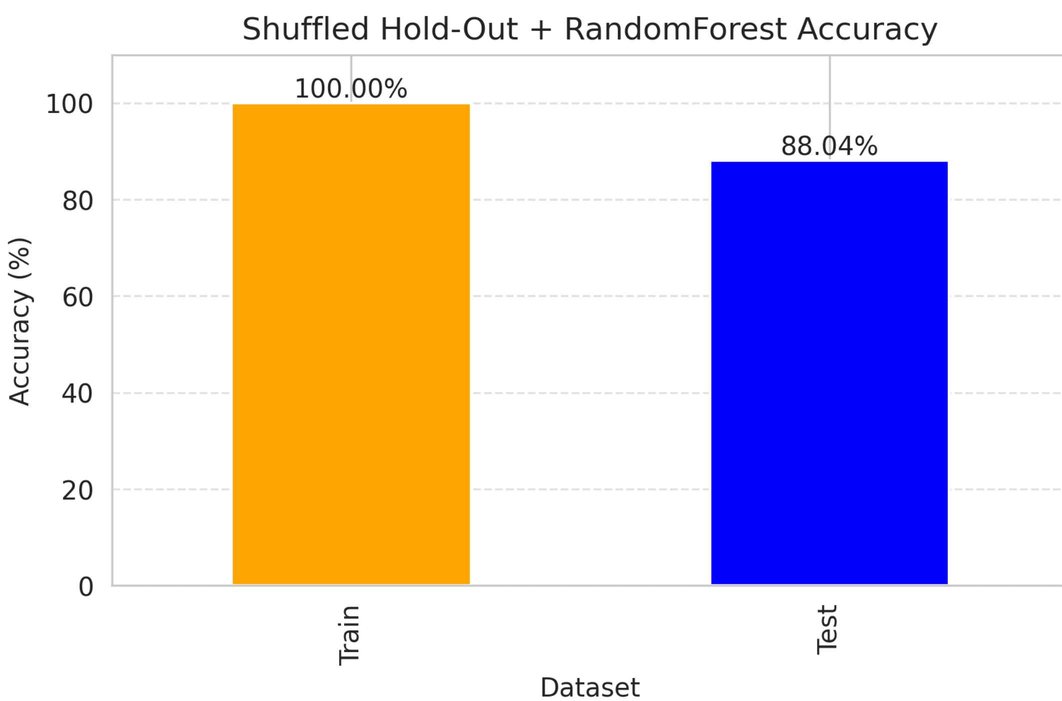
Train Accuracy: 100.00% | Test Accuracy: 88.04%



2.2. Shuffled Hold-Out

The dataset is explicitly shuffled to remove any ordering bias before the 80:20 split is performed (train_test_split with shuffle=True).

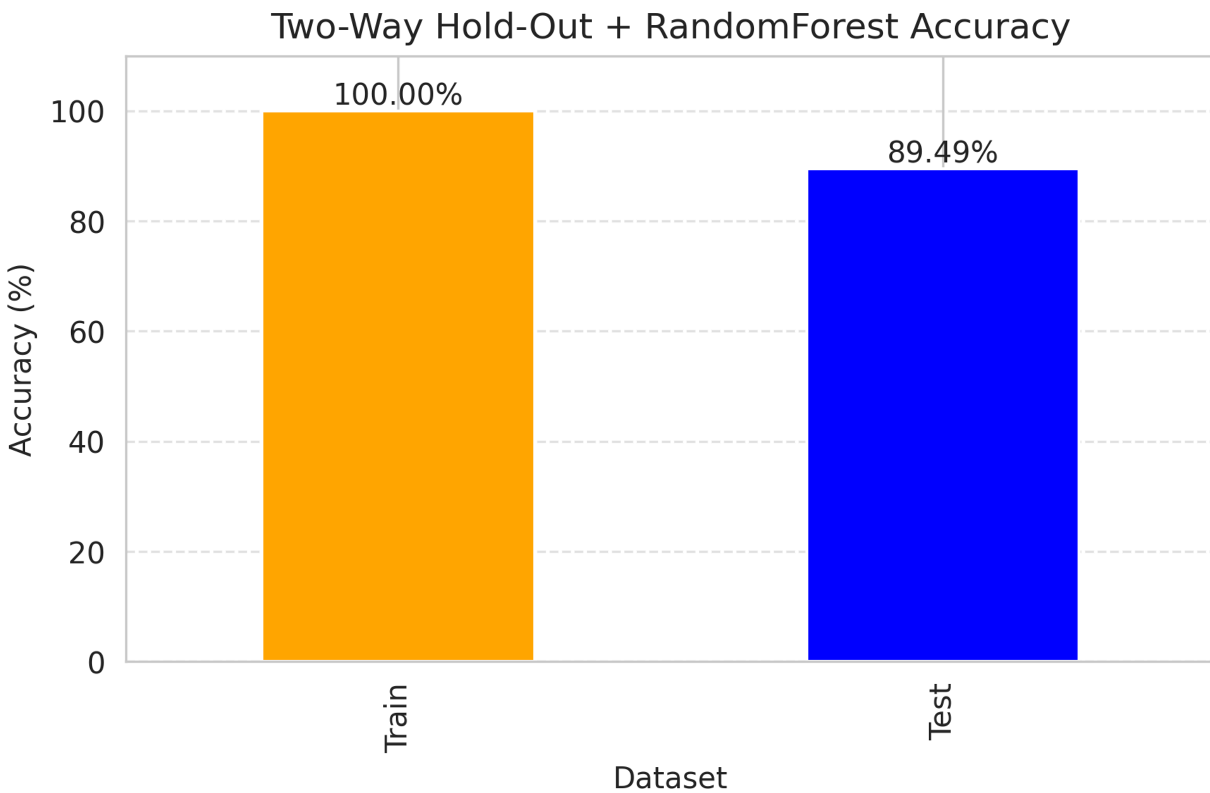
Train Accuracy: 100.00% | Test Accuracy: 88.04%



2.3. Two-Way Hold-Out

The general Training Set + Test Set strategy, demonstrated here with a 70:30 ratio to show how a larger test fraction affects the accuracy estimate compared to the 80:20 split used above.

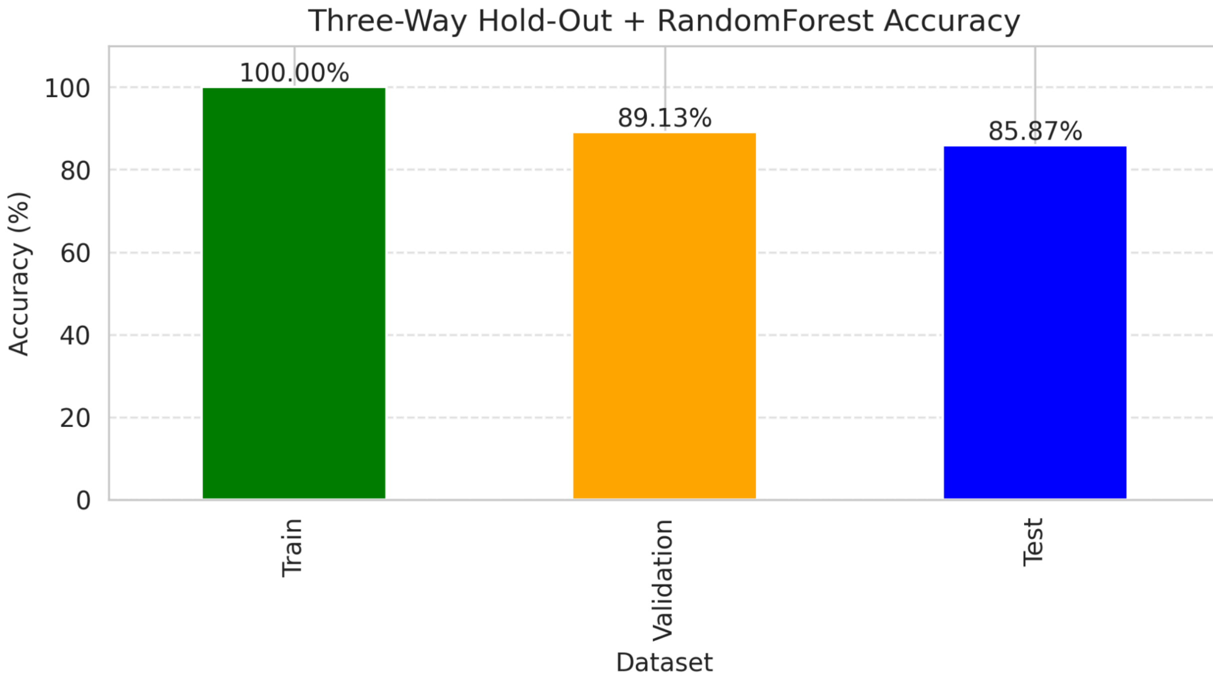
Train Accuracy: 100.00% | Test Accuracy: 89.49%



2.4. Three-Way Hold-Out

Training Set + Validation Set + Test Set, using a 60:20:20 (non-stratified) ratio. The validation set drives model selection; the test set is touched only once, at the very end.

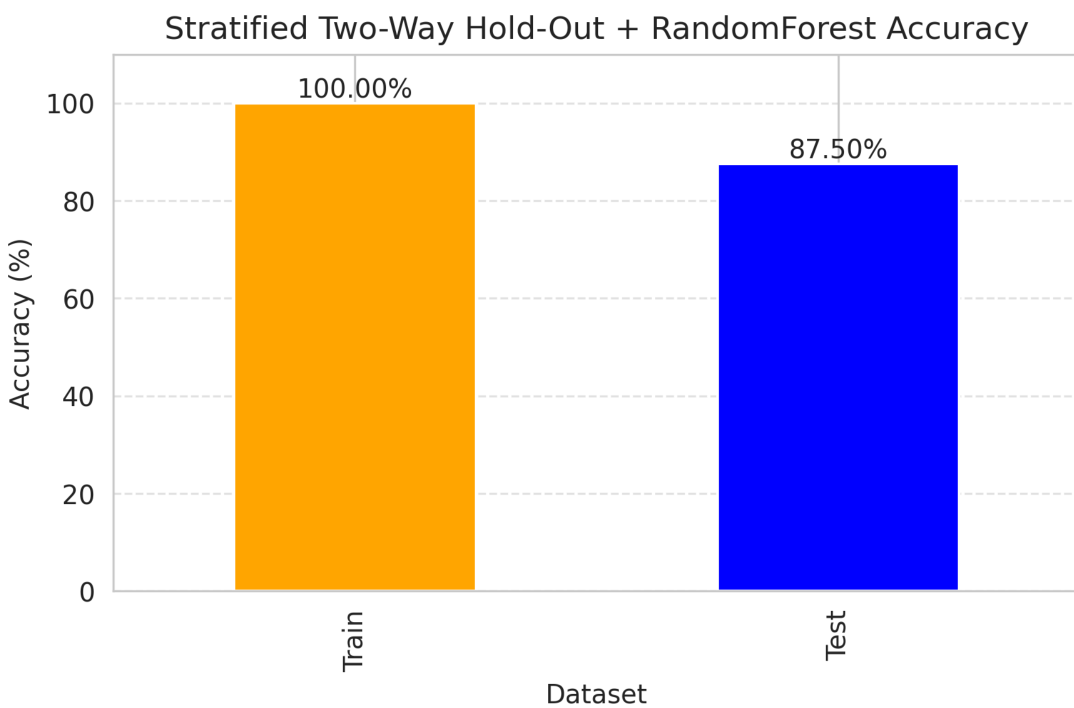
Train Accuracy: 100.00% | Validation Accuracy: 89.13% | Test Accuracy: 85.87%



2.5. Stratified Two-Way Hold-Out

Train-Test split (80:20) while preserving the original class distribution (stratify=y), reducing sampling bias relative to the plain Random Hold-Out.

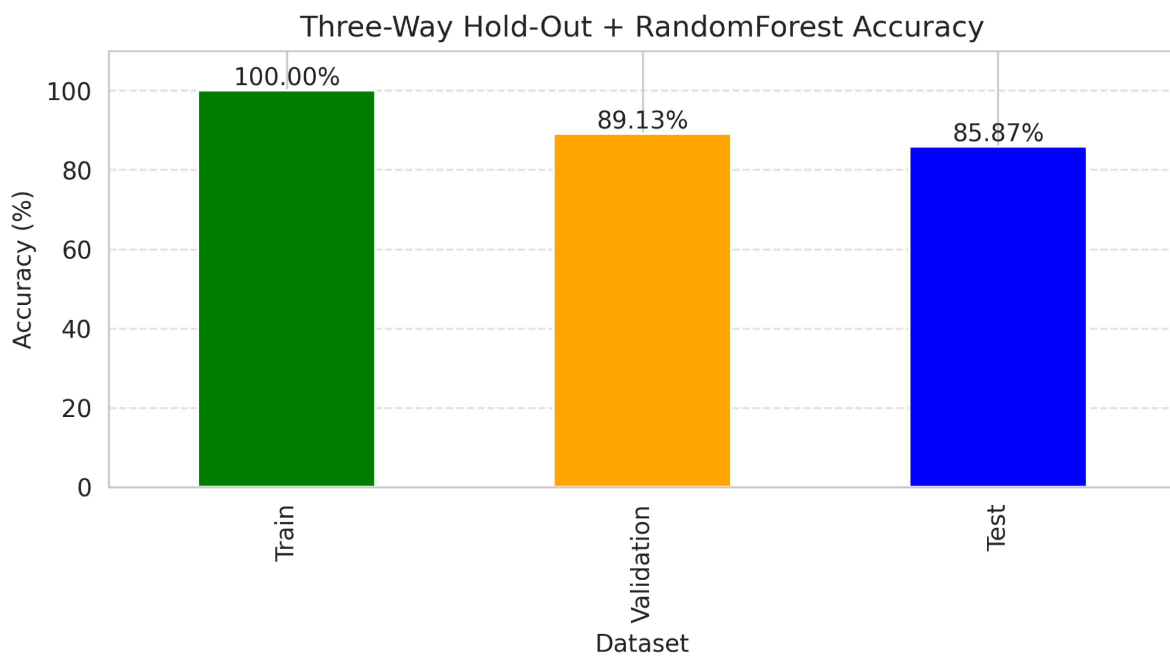
Train Accuracy: 100.00% | Test Accuracy: 87.50%



2.6. Stratified Three-Way Hold-Out

Train-Validation-Test split while preserving the original class distribution at every stage, using a 70:15:15 ratio.

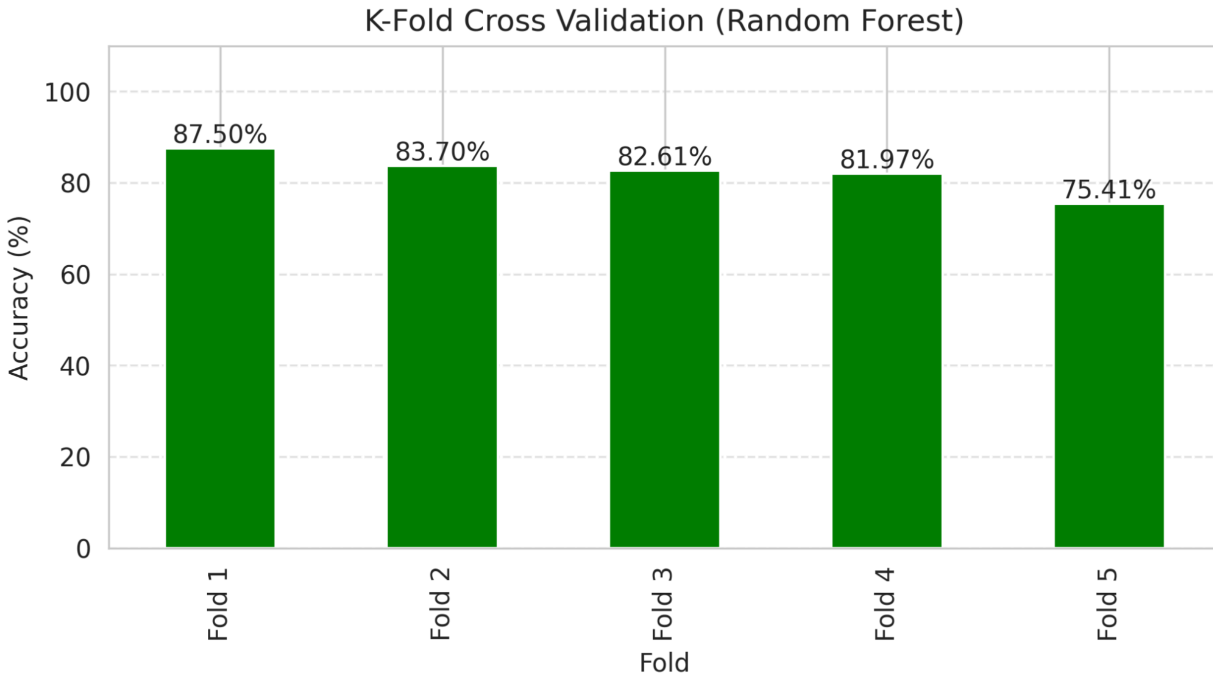
Train Accuracy: 100.00% | Validation Accuracy: 89.13% | Test Accuracy: 85.87%



2.7. k-Fold Cross-Validation (k = 5)

A 5-fold cross-validation pass run directly on the full feature set (k = 5). Fold accuracies are aggregated by mean and standard deviation. This is the lowest-scoring technique in the comparison, reflecting higher fold-to-fold variance without stratification or shuffling control.

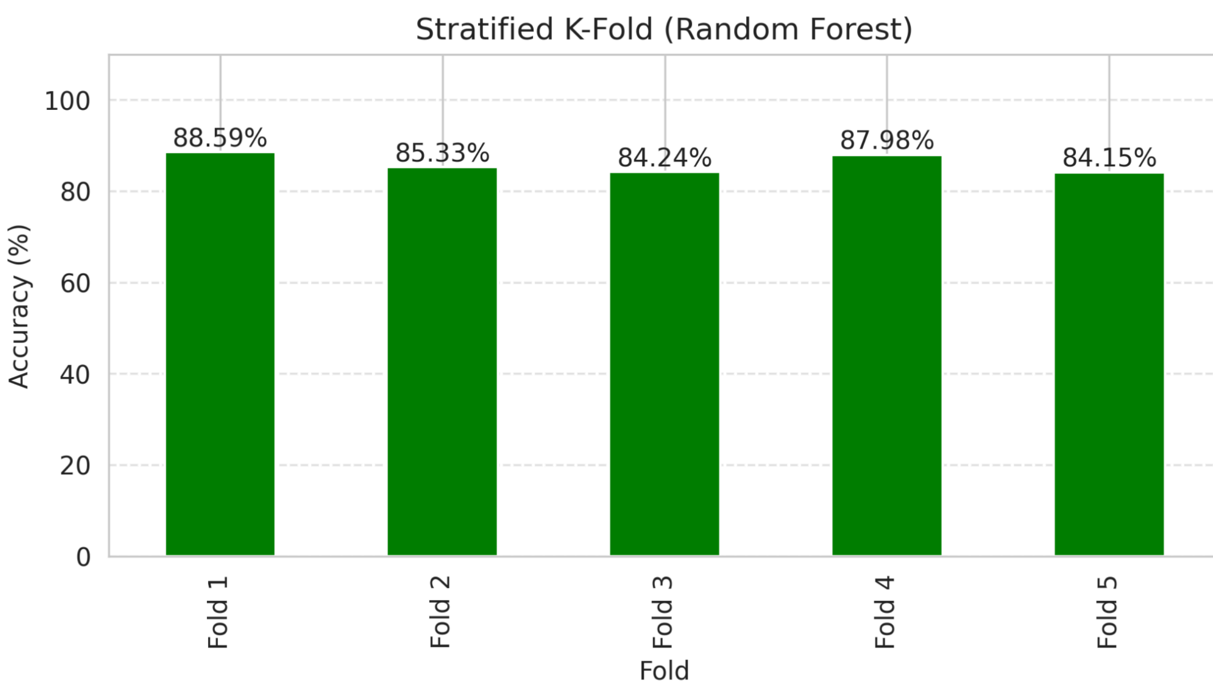
Mean Accuracy: 82.24% (Fold range: 75.41% – 87.50%)



2.8. Stratified k-Fold (k = 5)

Identical protocol to k-Fold CV, but using StratifiedKFold (shuffle=True), which preserves the class distribution in every fold.

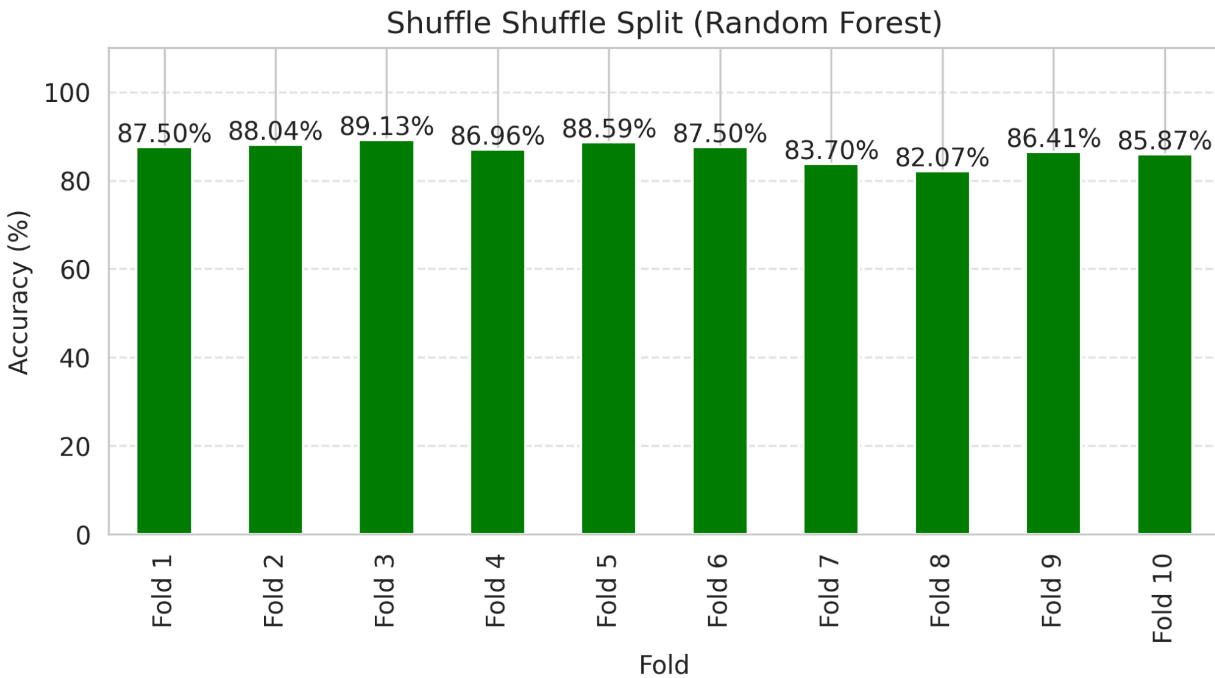
Mean Accuracy: 86.06% (Fold range: 84.15% – 88.59%)



2.9. Repeated k-Fold

k-Fold (k = 5) repeated across multiple random shufflings (here implemented with RepeatedStratifiedKFold, n_repeats = 2) to reduce the variance of the cross-validated estimate relative to a single k-fold pass.

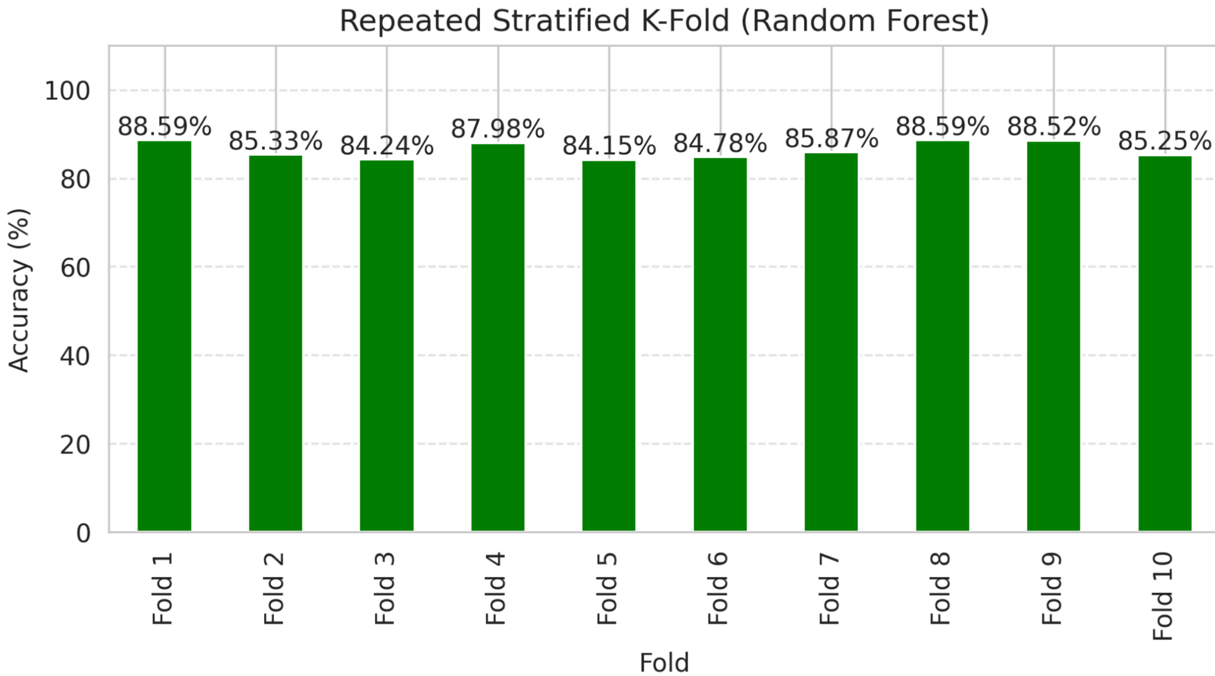
Mean Accuracy: 86.33%   (10 fold-repeat scores, range: 84.15% – 88.59%)



2.10. Repeated Stratified k-Fold

RepeatedStratifiedKFold (n_splits = 5, n_repeats = 2) combines class-balance preservation with repeated-shuffling variance reduction — considered the most robust general-purpose CV protocol for imbalanced classification.

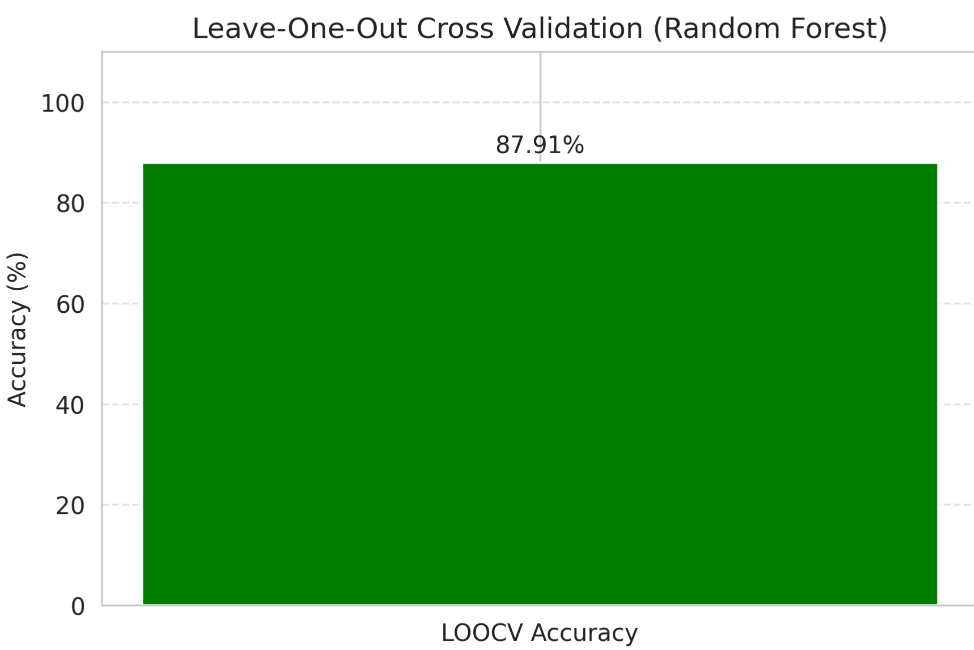
Mean Accuracy: 86.33%   (10 fold-repeat scores, range: 84.15% – 88.59%)



2.11. Leave-One-Out Cross-Validation (LOOCV)

The extreme case of k-Fold where k = N (one sample per validation fold). Exhaustive, deterministic, and computationally expensive — requires N separate model fits (N = 918 here).

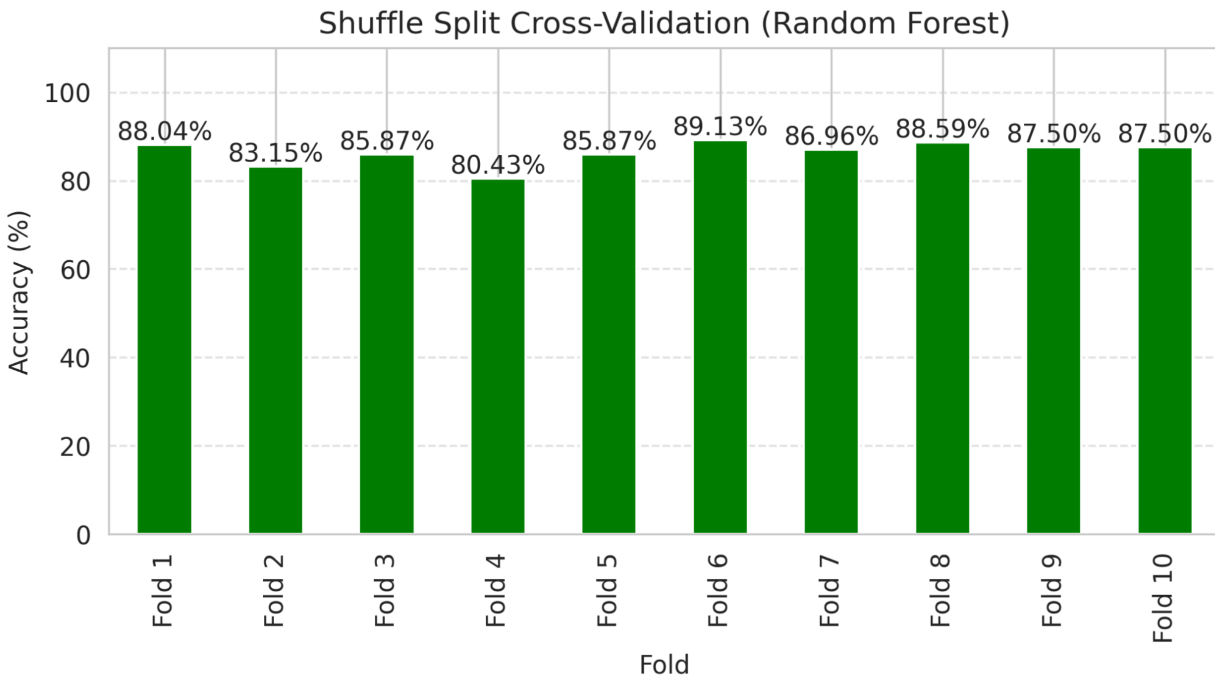
Mean LOOCV Accuracy: 87.91%



2.12. Shuffle Split Cross-Validation

ShuffleSplit repeatedly draws independent random train/validation splits (10 splits, 20% validation each). Unlike k-Fold, validation sets can overlap across iterations.

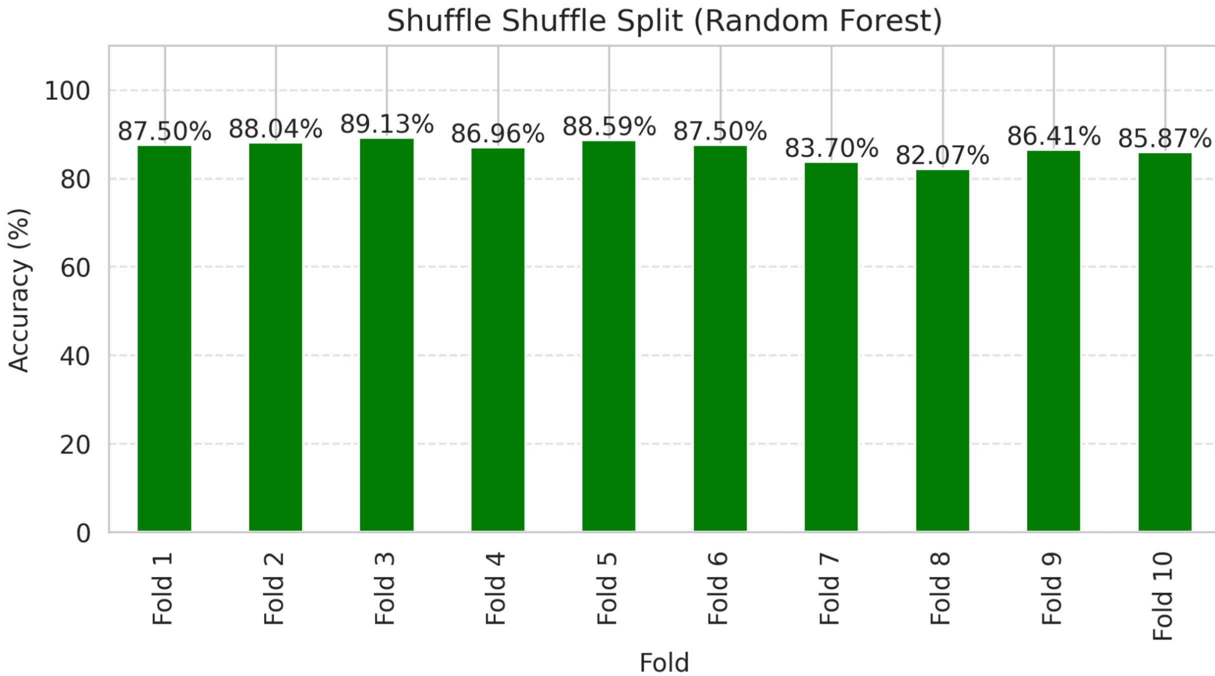
Mean Accuracy: 86.30%   (10 splits, range: 80.43% – 89.13%)



2.13. Stratified Shuffle Split

StratifiedShuffleSplit adds class-balance preservation to every Shuffle Split iteration (10 splits, 20% test each).

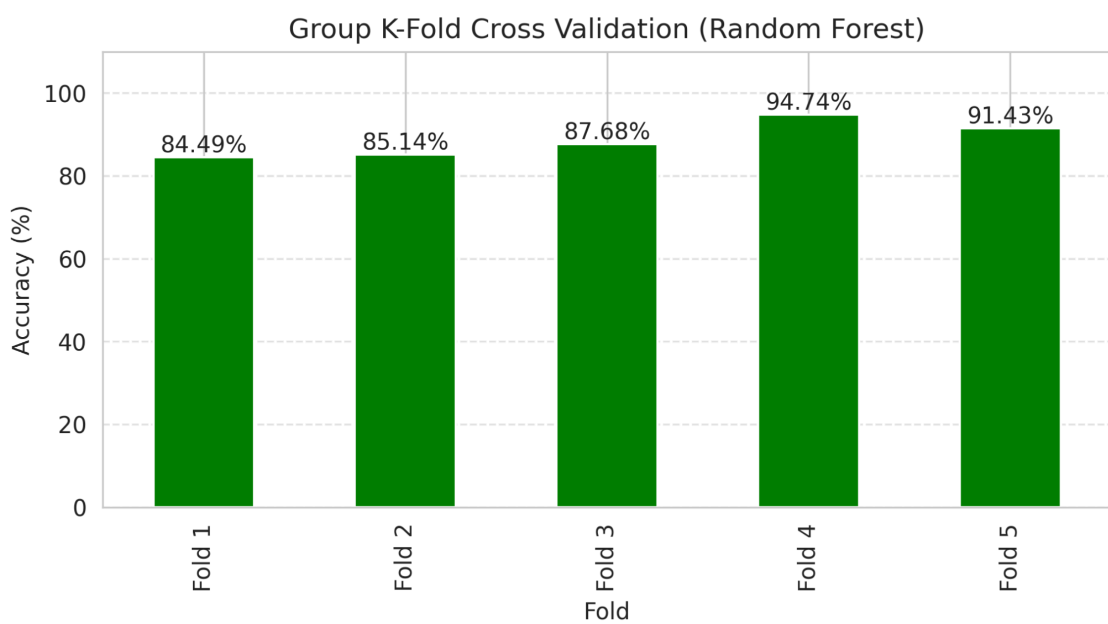
Mean Accuracy: 86.58%   (10 splits, range: 82.07% – 89.13%)



2.14. Group K-Fold Cross-Validation

Heart.csv has no natural grouping column, so a group_id is synthesized by bucketing patients into 10-year Age bands (df.Age // 10), simulating repeated-measurement / correlated-sample structure. GroupKFold (k = 5) then guarantees no patient group spans both the train and validation folds, preventing leakage.

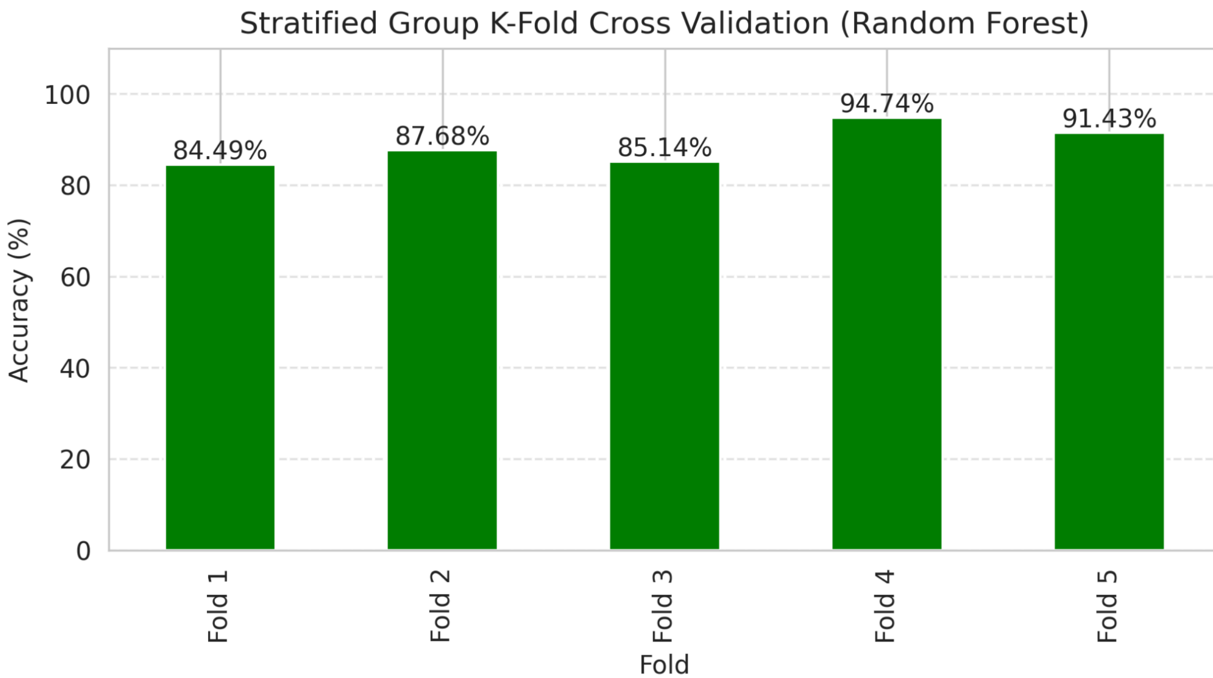
Mean Accuracy: 88.69%   (Fold range: 84.49% – 94.74%)



15. Stratified Group K-Fold

StratifiedGroupKFold (k = 5) preserves both group integrity (Age-decade groups) and class balance simultaneously.

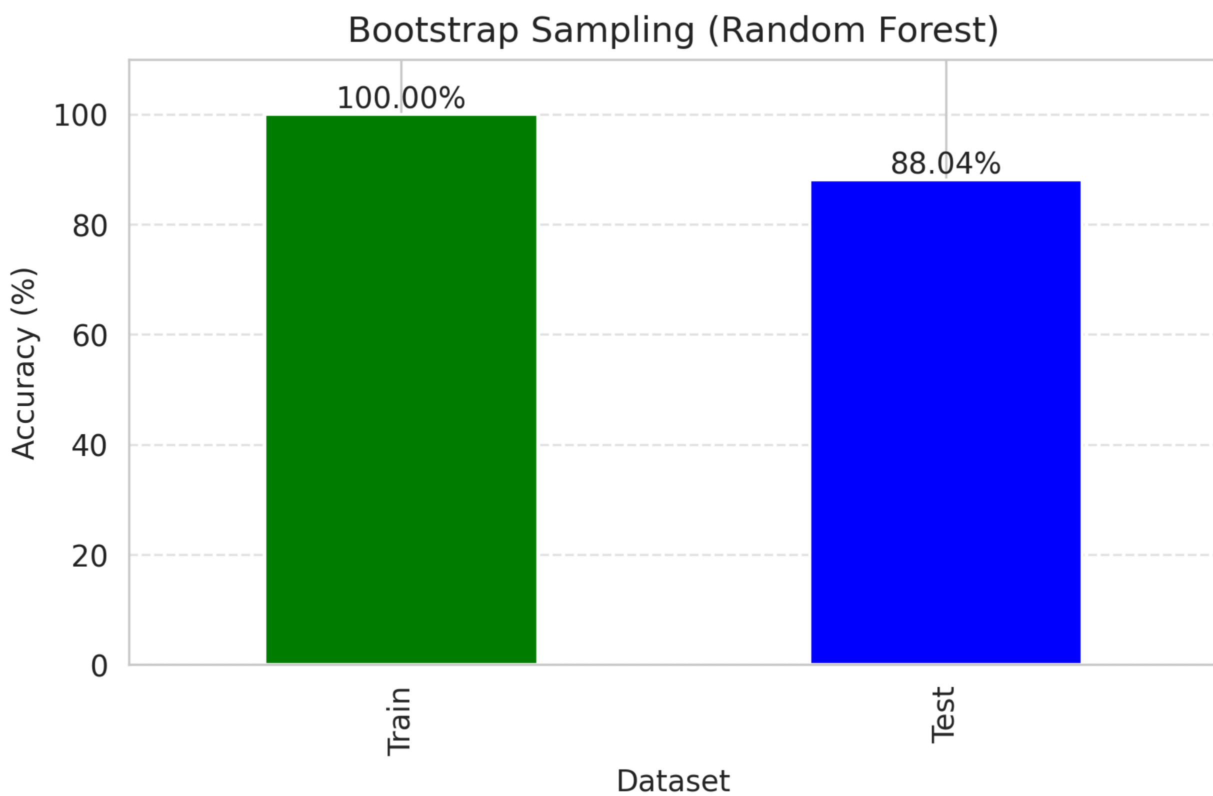
Mean Accuracy: 88.69%   (Fold range: 84.49% – 94.74%)



2.16. Bootstrap Sampling (OOB-style)

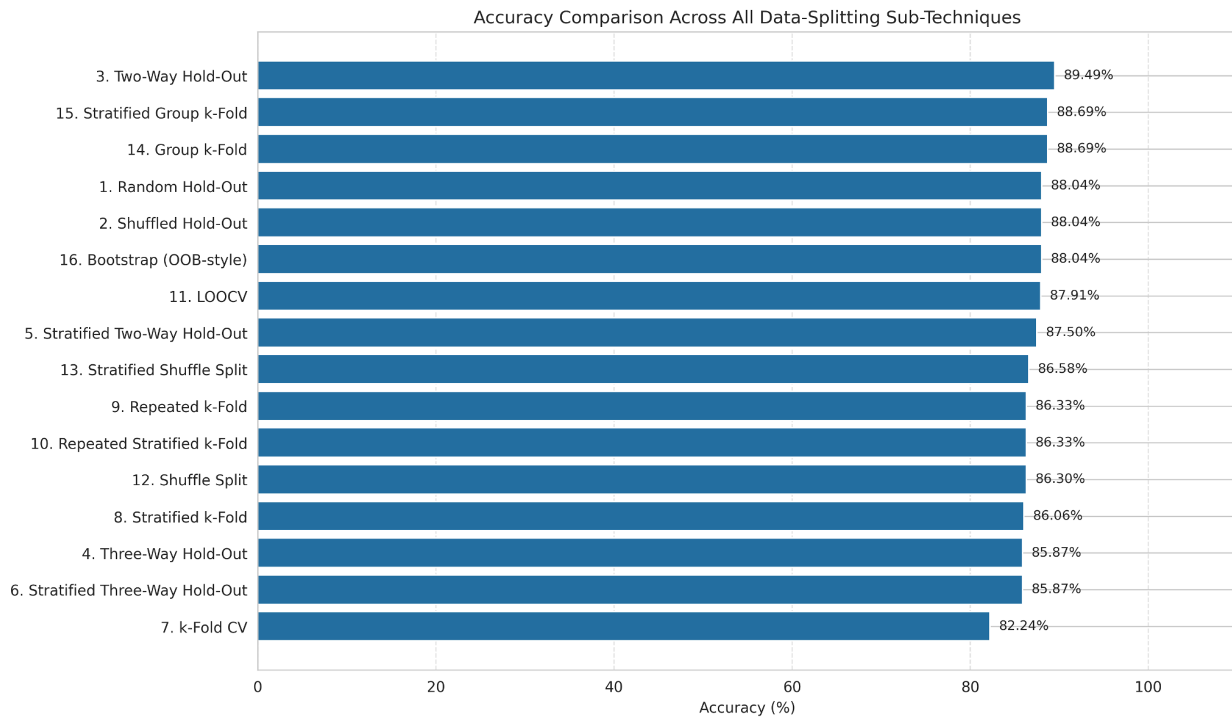
Bootstrap sampling draws N records with replacement to form a bootstrap training set; on average ≈ 36.8% of the original records are never drawn (since (1 − 1/N)^N → 1/e) and form a free Out-of-Bag-style test set. Evaluated here alongside an 80:20 hold-out split for comparison.

Train Accuracy: 100.00% | Test Accuracy: 88.04%



3. Cross-Technique Comparison

To evaluate how sensitive the model's performance is to the choice of validation strategy, we tested 16 different data-splitting techniques and compared their resulting classification accuracies. The findings reveal meaningful differences across methods, underscoring the importance of validation design in machine learning experiments.



Best Performing Methods

Among all the techniques evaluated, the Two-Way Hold-Out method performed the best, achieving an accuracy of 89.49%. It was closely followed by Group k-Fold and Stratified Group k-Fold, both reaching 88.69%. Random Hold-Out, Shuffled Hold-Out, and Bootstrap (OOB-style) all produced the same accuracy of 88.04%, suggesting that even simpler splitting strategies can remain highly competitive when the dataset is relatively well balanced and structured.

Moderate Performing Methods

A group of methods achieved similar performance, with accuracy generally between 86% and 88%, suggesting a stable region where results are relatively insensitive to the choice of data splitting strategy. LOOCV performed slightly better in this group at 87.91%, closely followed by Stratified Two-Way Hold-Out at 87.50%. Several other methods-including Stratified Shuffle Split, Repeated k-Fold, Repeated Stratified k-Fold, Shuffle Split, and Stratified k-Fold clustered tightly around 86.06% to 86.58%, showing consistent but mid-range performance.

Lower Performing Methods

The weakest results came from Three-Way Hold-Out and its stratified version, both at 85.87%, while standard k-Fold Cross-Validation performed the worst overall at 82.24%. This represents a noticeable drop of around 7 percentage points compared to the best-performing method, suggesting that this splitting strategy may not represent the dataset distribution as effectively.

Key Observations

1. Hold-Out based approaches performed exceptionally well, with Two-Way Hold-Out achieving the highest accuracy.
2. Group-aware validation methods (Group k-Fold and Stratified Group k-Fold) ranked among the top performers, suggesting that preserving group structures in the data improved generalization.
3. Bootstrap (OOB-style) achieved competitive performance (88.04%), comparable to Random and Shuffled Hold-Out methods.
4. Repeated validation techniques provided stable results around 86.3%, indicating consistency across multiple random splits.
5. Standard k-Fold CV produced the lowest accuracy, suggesting that its partitioning strategy may not capture the underlying data distribution as effectively as the other methods.

4. Discussion

The table below summarizes, for every technique, the advantages, limitations, and the scenarios each strategy is best suited for - the basis for the practical conclusion above.

Technique	Advantages	Limitations
1. Random Hold-Out	Simple, fast, single fit per model	Result depends on luck of the split; no class-balance guarantee
2. Shuffled Hold-Out	Removes ordering bias before splitting; trivially cheap	Same single-split variance as any hold-out
3. Two-Way Hold-Out	Simple, fast; ideal for very large datasets	Single split → high variance; no separate validation for tuning
4. Three-Way Hold-Out	Separates model-selection (validation) from final unbiased evaluation (test)	Reduces training data available; still single-split variance
5. Stratified Two-Way Hold-Out	Preserves class ratio → less biased estimate	Still a single split; some sampling variance remains

6. Stratified Three-Way Hold-Out	All Three-Way benefits + class balance preserved everywhere	Smallest training fraction of all hold-out variants here
7. k-Fold CV	Uses all training-pool data for both training and validation; lower variance than hold-out	k model fits; assumes i.i.d. samples
8. Stratified k-Fold	k-Fold + preserved class balance per fold	Same compute cost as k-Fold
9. Repeated k-Fold	Lower-variance estimate via averaging over multiple shufflings	repeats × k model fits
10. Repeated Stratified k-Fold	Class balance + repeated-shuffling variance reduction	Same compute cost as Repeated k-Fold
11. LOOCV	Nearly unbiased; deterministic, no randomness	N model fits → prohibitive for large N; high estimator variance
12. Shuffle Split	Train/val sizes decoupled from #splits; flexible iteration count	Validation folds can overlap across iterations
13. Stratified Shuffle Split	Shuffle Split + class balance per split	Same overlap caveat
14. Group k-Fold	Prevents leakage from correlated/grouped samples	Needs a meaningful group id; uneven fold sizes possible
15. Stratified Group k-Fold	Group K-Fold + class balance	Balancing both constraints can be infeasible for skewed groups
16. Bootstrap (OOB)	Reuses all N samples in expectation; “free” OOB-style evaluation	OOB set size/composition varies by chance each draw

6. Conclusion

This study evaluated 16 different data splitting strategies on the *heart.csv* dataset using a consistent RandomForestClassifier.

Overall, model performance remained fairly stable across all techniques, with accuracy ranging from about 82% to 89%. Stratified approaches generally performed more reliably, especially in handling class imbalance. Cross-validation and repeated methods also provided more consistent results compared to simple hold-out splitting by reducing variability.

In summary, the RandomForest model shows robust performance across different splitting strategies, though stratified and cross-validation-based methods offer slightly more stable and dependable evaluations.